Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 19: Approved Solutions

One-to-One and Onto Functions

Covers Chartrand & Zhang, Section 5.4

Status of this set

Ungraded practice. Solutions released with it. Write proofs in complete sentences.

Part I: How to Practice This Material

Four jobs, four templates

Know which one you are doing

	To prove	To disprove
One-to-one	Assume $f(a) = f(b)$; derive $a = b$.	Name $a \neq b$ with $f(a) = f(b)$.
Onto	Take arbitrary b; produce a with $f(a) = b$.	Name a b with no preimage.

Scratch work runs backwards

To prove onto you must *produce* a preimage. Find it by solving $f(x) = b$ for x on scratch paper, then write the proof forwards: "Let b be given. Set $a = \ldots$ Then $f(a) = b$."
The finished proof hides the search. That is normal, and expected of you too.

Three mistakes to expect

1. **Starting a one-to-one proof from $a \neq b$.** Use the contrapositive form.
2. **Asserting onto without producing a preimage.** "Every value is hit" is a claim, not a proof.
3. **Ignoring the codomain.** The same rule can be onto or not depending on it.

Part II: Practice Problems

Section A — Explicit functions

1. Let $A = \{1, 2, 3\}$ and $B = \{p, q, r, s\}$. For each function, state whether it is one-to-one, onto, both, or neither, with witnesses for failures.

(a) $f = \{(1, p), (2, q), (3, r)\}$

(b) $g = \{(1, p), (2, p), (3, s)\}$

(c) $h : B \to A$ with $h = \{(p, 1), (q, 2), (r, 3), (s, 1)\}$

Approved solution

(a) One-to-one, not onto. Distinct inputs have distinct images. Not onto: s is the image of nothing.
(b) Neither. Not one-to-one: $g(1) = g(2) = p$. Not onto: q and r are images of nothing.
(c) Onto, not one-to-one. Every element of A is hit: $1 = h(p)$, $2 = h(q)$, $3 = h(r)$. Not one-to-one: $h(p) = h(s) = 1$.

The pattern is forced by size. With $|A| = 3 < 4 = |B|$, no function $A \to B$ can be onto, and no function $B \to A$ can be one-to-one. That is Result (5.3) and its contrapositive — the pigeonhole principle in miniature.

Section B — Proving

2. Prove that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 7x + 2$ is bijective, and find f^{-1}.

Approved solution

One-to-one. Assume $f(a) = f(b)$, so $7a + 2 = 7b + 2$. Subtracting 2 gives $7a = 7b$, and dividing by 7 gives $a = b$. Hence f is one-to-one.
Onto. Let $b \in \mathbb{R}$. Set $a = \dfrac{b - 2}{7}$, which is a real number. Then

$$f(a) = 7\left(\frac{b-2}{7}\right) + 2 = (b - 2) + 2 = b.$$

Hence f is onto. Therefore f is bijective. ■
Inverse. $f^{-1}(y) = \dfrac{y - 2}{7}$.

The preimage in the onto half came from solving $7x + 2 = b$ — the same computation that produces the inverse. **Proving a linear function onto and finding its inverse are the same algebra**, done once and reported twice.

3. For each function, determine whether it is one-to-one and whether it is onto. Justify.

(a) $f : \mathbb{Z} \to \mathbb{Z}$, $f(n) = n + 5$.

(b) $g : \mathbb{Z} \to \mathbb{Z}$, $g(n) = 3n$.

(c) $h : \mathbb{R} \to \mathbb{R}$, $h(x) = x^3 - x$.

Approved solution

(a) Both. One-to-one: $a+5=b+5$ gives $a=b$. Onto: given $b\in\mathbb{Z}$, take $a=b-5$, an integer, and $f(a)=b$.
(b) One-to-one, not onto. $3a=3b$ gives $a=b$. But 1 has no preimage: $3n=1$ forces $n=\frac{1}{3}$, which is not an integer.
(c) Not one-to-one; onto. Not one-to-one: $h(0)=0$, $h(1)=0$, and $h(-1)=0$, so three distinct inputs share an image. Onto: h is a cubic polynomial, continuous, with $h(x)\to\pm\infty$ as $x\to\pm\infty$, so by the intermediate value theorem every real number is attained.

Compare (a) and (b). Both are one-to-one, and the domain and codomain are the same in each. Only (a) is onto — "multiply by 3" cannot reach the non-multiples of 3, while "add 5" can be undone within $\mathbb{Z}$.
Part (c)'s witnesses come from factoring: $x^3-x=x(x-1)(x+1)$, which is zero at three points. Factoring produces counterexamples; guessing rarely does.

Section C — The codomain matters

4. Consider the rule $f(x)=x^2$ in three settings. For each, state whether it is one-to-one and whether it is onto.

(a) $f:\mathbb{R}\to\mathbb{R}$

(b) $f:\mathbb{R}\to[0,\infty)$

(c) $f:[0,\infty)\to[0,\infty)$

Approved solution

(a) Neither. Not one-to-one: $f(1)=f(-1)=1$. Not onto: -1 has no preimage, since $x^2\geq 0$.
(b) Not one-to-one; onto. Same witness as before for the first. Onto: given $b\geq 0$, take $a=\sqrt{b}$, and $f(a)=b$.
(c) Both — bijective. One-to-one: if $a^2=b^2$ with $a,b\geq 0$, taking nonnegative square roots gives $a=b$. Onto: as in (b).

One rule, three functions, three answers. Restricting the codomain fixed ontoness; restricting the domain fixed injectivity. **Neither property belongs to the formula** — they belong to the function, which includes both sets.
This is the single most important idea in Section 5.4, and it is why "is x^2 one-to-one?" is not a well-posed question.

Section D — Writing

5. In five to eight complete sentences, explain why the standard way to prove a function is one-to-one starts from $f(a)=f(b)$ rather than from $a\neq b$. Name the result from earlier in the term that justifies the switch, and say what practical advantage it gives.

Approved solution

Model answer. The definition says that if $a\neq b$ then $f(a)\neq f(b)$, but its contrapositive says that if $f(a)=f(b)$ then $a=b$, and by Theorem 1.46 an implication and its contrapositive are logically

equivalent. So proving either one proves the other, and we are free to choose whichever is easier to work with. The second version is much easier, because it hands you an *equation* as a hypothesis. From $f(a) = f(b)$ you can substitute the formula and manipulate both sides algebraically until $a = b$ falls out, as in $5a - 3 = 5b - 3$. By contrast, assuming $a \neq b$ gives you an inequality, and inequalities cannot be simplified in the same way — there is very little you can do with "these two things are different." So the switch is not a matter of taste; it converts an unusable hypothesis into a usable one.

Full credit requires naming the contrapositive and explaining the *practical* advantage. An answer that only says "they're equivalent" earns about half.

Section E — Challenge (optional)

6. Let A be a finite set and $f : A \to A$ a function. Prove that f is one-to-one if and only if f is onto. Then give an example showing this fails for infinite sets.

Approved solution

($\Rightarrow$) Suppose f is one-to-one and $|A| = n$. Distinct inputs have distinct images, so the range $f(A)$ contains n elements. But $f(A) \subseteq A$ and $|A| = n$, so $f(A) = A$ — a subset of a finite set with the same size must be the whole set. Hence f is onto.
($\Leftarrow$) Suppose f is onto, so $f(A) = A$ and the range has n elements. Those n images come from n inputs, so no two inputs can share an image — otherwise fewer than n distinct values would be produced. Hence f is one-to-one.
Failure for infinite sets. Take $A = \mathbb{Z}$ and $f(n) = 2n$. This is one-to-one ($2a = 2b$ gives $a = b$) but not onto (1 has no preimage). So the two properties come apart.

Both directions leaned on finiteness, in the step where a subset of the same size must be the whole set. That step is exactly what fails for $\mathbb{Z}$ — an infinite set can have a proper subset of the "same size," which is the definition of infinite and the starting point for countability.

Instructor notes

- **Problem 4 is the set's centrepiece.** One rule, three domain/codomain pairs, three different answers. Put all three on the board at once — it makes "the codomain is part of the function" impossible to miss.
- **Problem 2's closing remark** — proving onto and finding the inverse are the same algebra — saves students real time on Homework 8.
- **Problem 3(c)** needs factoring to find the witnesses. Students who hunt at random will not find three roots.
- **Problem 6** previews cardinality: the finite proof breaks exactly where infinite sets misbehave. Worth flagging even if you skip the challenge.
- **Problem 1's closing remark** is the pigeonhole principle appearing early.
- Reserved for Homework 8: Section 5.4 even-numbered exercises.